ELEC50011 PART B: PROBABILITY AND STATISTICS – SOLUTIONS

Notation:
[X-B] Indicates X points given for a "Bookwork" answer.
[X-E] Indicates X points given for a "New Example" answer.
[X-A] Indicates X points given for a "New Application" answer.

1. Consider two continuous random variables X and Y that have the joint probability density function (pdf) given in the following equation:

$$f_{X,Y}(x,y) = \begin{cases} x^2y + ax^2 + by + c, & \text{if } 0 < x < 1 \text{ and } 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $a, b, c \geq 0$ are non-negative constants.

a) What relationship must a, b and c satisfy to ensure that $f_{X,Y}(x,y)$ describes a valid pdf? [3]

SOLUTION. Since the constants and random variables are constrained to be non-negative, it is necessarily the case that the pdf is non-negative. We hence only need to ensure that the pdf integrates to one. [1-B]

To this end: [2-E]

$$\begin{aligned}
&\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy \\
&= \int_0^1 \int_0^1 \left(x^2y + ax^2 + by + c\right) dxdy \\
&= \int_0^1 \left[\frac{1}{3}x^3y + \frac{a}{3}x^3 + bxy + cx\right]_0^1 dy \\
&= \int_0^1 \left(\frac{1}{3}y + \frac{a}{3} + by + c\right) dy \\
&= \int_0^1 \left(\frac{1+3b}{3}y + \frac{a+3c}{3}\right) dy \\
&= \left[\frac{2+3b}{12}y^2 + \frac{a+3c}{3}y\right]_0^1 \\
&= \frac{1+3b}{6} + \frac{a+3c}{3} \\
&= \frac{2a+3b+6c+1}{6} \\
&\stackrel{!}{=} 1
\end{aligned}$$

Hence:

$$2a + 3b + 6c = 5$$

b) What additional condition on a, b and c ensures that X and Y are independent? [3]

SOLUTION. We need to find a parameterisation that allows for the factorisation $f_{X,Y}(x,y) = f_X(x)f_Y(y)$. [1-B]

We can do this by completing the square: [1-E]

$$x^2y + ax^2 + by + c = (x^2 + b)(y + a) - ab + c$$

Hence, as long as $ab = c$, we have:

$$f_{X,Y}(x,y) = (x^2 + b)(y + a) = f_X(x) f_Y(y)$$

with: [1-E]

$$f_X(x) = x^2 + b$$
$$f_Y(y) = y + a$$

For the remainder of Question 1, we set $a = 1$, $b = 1$ and $c = 0$.

c) Find the probability that $Y \geq X$, i.e., $P(Y \geq X)$. [3]

SOLUTION. We have: [3-E]

$$\begin{aligned}
P(Y \geq X) &= \int_0^1 \int_0^x (x^2y + x^2 + y)\,dy\,dx \\
&= \int_0^1 \left[\frac{1}{2}x^2y^2 + x^2y + \frac{1}{2}y^2\right]_0^x dx \\
&= \int_0^1 \left(\frac{1}{2}x^4 + x^3 + \frac{1}{2}x^2\right) dx \\
&= \left[\frac{1}{10}x^5 + \frac{1}{4}x^4 + \frac{1}{6}x^3\right]_0^1 \\
&= \frac{1}{10} + \frac{1}{4} + \frac{1}{6} \\
&= \frac{12}{120} + \frac{30}{120} + \frac{20}{120} \\
&= \frac{62}{120} \\
&= \frac{31}{60}
\end{aligned}$$

d) Find the marginals $f_X(x)$ and $f_Y(y)$. [3]

SOLUTION. For the marginal of X, we integrate over y: [2-E]

$$\begin{aligned}
f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy \\
&= \int_0^1 \left(x^2y + x^2 + y\right) dy \\
&= \left[\frac{1}{2}x^2y^2 + x^2y + \frac{1}{2}y^2\right]_0^1 \\
&= \frac{3}{2}x^2 + \frac{1}{2}
\end{aligned}$$

For the marginal of Y, we integrate over x: [1-E]

$$\begin{aligned}
f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx \\
&= \int_0^1 \left(x^2y + x^2 + y\right) dx \\
&= \left[\frac{1}{3}x^3y + \frac{1}{3}x^3 + xy\right]_0^1 \\
&= \frac{4}{3}y + \frac{1}{3}
\end{aligned}$$

e) Compute the expected values of X and Y, i.e., $E(X)$ and $E(Y)$. [3]

SOLUTION. Computing the expectation of X: [2-E]

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^1 \left(\frac{3}{2}x^3 + \frac{1}{2}x\right) dx \\ &= \left[\frac{3}{8}x^4 + \frac{1}{4}x^2\right]_0^1 \\ &= \frac{5}{8} \end{aligned}$$

Computing the expectation of Y: [1-E]

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \int_0^1 \left(\frac{4}{3}y^2 + \frac{1}{3}y\right) dy \\ &= \left[\frac{4}{9}y^3 + \frac{1}{6}y^2\right]_0^1 \\ &= \frac{11}{18} \end{aligned}$$

f) Compute the variances of X and Y, i.e., $Var(X)$ and $Var(Y)$. [3]

SOLUTION. We proceed in a manner similar to the the computation of the expected value in the previous question. For the variance of X, we first compute: [1-E]

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \int_0^1 \left(\frac{3}{2}x^4 + \frac{1}{2}x^2\right) dx \\ &= \left[\frac{3}{10}x^5 + \frac{1}{6}x^3\right]_0^1 \\ &= \frac{14}{30} = \frac{7}{15} \end{aligned}$$

Then: [1-B]

$$Var(X) = E(X^2) - (E(X))^2 = \frac{7}{15} - \left(\frac{5}{8}\right)^2 = \frac{73}{960} \approx 0.076$$

Analogously, for Y:

$$\begin{aligned} E(Y^2) &= \int_{-\infty}^{\infty} y^2 f_Y(y) dy \\ &= \int_0^1 \left(\frac{4}{3}y^3 + \frac{1}{3}y^2\right) dy \\ &= \left[\frac{1}{3}y^4 + \frac{1}{9}y^3\right]_0^1 \\ &= \frac{4}{9} \end{aligned}$$

Then: [1-E]

$$Var(Y) = E(Y^2) - (E(Y))^2 = \frac{4}{9} - \left(\frac{11}{18}\right)^2 = \frac{23}{324} \approx 0.071$$

g) Find the covariance between X and Y, i.e., $Cov(X,Y)$, as well as their correlation, i.e., $Corr(X,Y)$. [4]

SOLUTION. We begin by computing: [2-E]

$$\begin{aligned}
E(XY) &= \int_0^1 \int_0^1 xy(x^2y+x^2+y)dydx \\
&= \int_0^1 \int_0^1 (x^3y^2+x^3y+xy^2)dydx \\
&= \int_0^1 \left[\frac{1}{3}x^3y^3+\frac{1}{2}x^3y^2+\frac{1}{3}xy^3\right]_0^1 dx \\
&= \int_0^1 \left(\frac{1}{3}x^3+\frac{1}{2}x^3+\frac{1}{3}x\right)dx \\
&= \left[\frac{1}{12}x^4+\frac{1}{8}x^4+\frac{1}{6}x^2\right]_0^1 \\
&= \frac{9}{24} = \frac{3}{8}
\end{aligned}$$

Then, using the results from the previous part: [1-B]

$$Cov(X,Y) = E(XY) - E(X)E(Y) = \frac{3}{8} - \frac{5}{8} \times \frac{11}{18} = -\frac{1}{144}$$

Finally, we can compute the correlation by using the formula from lectures: [1-B]

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{-\frac{1}{144}}{\sqrt{\frac{73}{960} \times \frac{23}{324}}} \approx -0.000510$$

We see there is a very weak negative correlation.

h) Find the conditional probability density function of Y given $X = x$. [2]

SOLUTION. We have: [2-E]

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{x^2y+x^2+y}{\frac{3}{2}x^2+\frac{1}{2}}$$

i) Compute the conditional expectation of Y given $X = x$. [3]

SOLUTION. The definition of the conditional expectation: [1-E]

$$E(Y|X=x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)dy$$

Then: [2-E]

$$\begin{aligned}
E(Y|X=x) &= \int_0^1 y\frac{x^2y+x^2+y}{\frac{3}{2}x^2+\frac{1}{2}}dy \\
&= \frac{1}{\frac{3}{2}x^2+\frac{1}{2}} \int_0^1 \left(x^2y^2+x^2y+y^2\right)dy \\
&= \frac{1}{\frac{3}{2}x^2+\frac{1}{2}} \left[\frac{1}{3}x^2y^3+\frac{1}{2}x^2y^2+\frac{1}{3}y^3\right]_0^1 \\
&= \frac{\frac{5}{6}x^2+\frac{1}{3}}{\frac{3}{2}x^2+\frac{1}{2}} = \frac{5x^2+2}{9x^2+3}
\end{aligned}$$

j) Evaluate $E(E(Y|X))$ using the result of part i). [3]

SOLUTION. To compute the double expectation, we need: [1-B]

$$E(E(Y|X)) = \int_{-\infty}^{\infty} E(Y|X=x) f_X(x) dx$$

Thus, have: [2-E]

$$\begin{aligned} E(E(Y|X)) &= \int_{-\infty}^{\infty} E(Y|X=x) f_X(x) dx \\ &= \int_0^1 \frac{\frac{5}{6}x^2 + \frac{1}{3}}{\frac{3}{2}x^2 + \frac{1}{2}} \left(\frac{3}{2}x^2 + \frac{1}{2}\right) dx \\ &= \int_0^1 \left(\frac{5}{6}x^2 + \frac{1}{3}\right) dx \\ &= \left[\frac{5}{18}x^3 + \frac{1}{3}x\right]_0^1 \\ &= \frac{5}{18} + \frac{1}{3} \\ &= \frac{11}{18} \end{aligned}$$

The answer matches the result computed earlier.

2. Consider a random variable $V \sim \mathcal{N}\left(0, \sigma_v^2\right)$, which is normally distributed with mean 0 and variance σ_v^2. We construct a new random variable, given by:

$$X = a\theta + bV$$

where $a \neq 0$ and b are known constants, and θ is an unknown parameter.

a) Find the probability density function $f_X(x)$ of X. [5]

SOLUTION. The equation describes a linear transformation of a Gaussian random variable, which itself is Gaussian. [1-B]

This means that we just need to compute the mean and variance of X. For the mean, we find: [1-B]

$$E(X) = a\theta + bE(V) = a\theta$$

For the variance: [1-B]

$$Var(X) = b^2 Var(V) = b^2\sigma_v^2$$

Since we know X to be Gaussian, it follows that: [2-B]

$$f_X(x) = \frac{1}{\sqrt{2\pi b^2 \sigma^2}} e^{-\frac{(x-a\theta)^2}{2b^2\sigma_v^2}}$$

Alternatively, the pdf can be found using the change-of-variable formula for a scalar function of a scalar random variable, but this is more involved.

b) We collect an independently and identically distributed (i.i.d.) sample $X_1, \ldots, X_n$ of size n. Compute the method-of-moments estimate for θ. [3]

SOLUTION. The method-of-moments estimate is obtained by equating a suitable moment to its sample estimate. In this case, $E(X) = a\theta$ is a simple function of θ, which is the quantity we care about. [1-E]

In this case: [2-E]

$$a\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i \Longleftrightarrow \widehat{\theta} = \frac{\frac{1}{n}\sum_{i=1}^{n} x_i}{a}$$

c) We collect an independently and identically distributed (i.i.d.) sample $X_1, \ldots, X_n$ of size n. Determine an expression for the maximum likelihood estimate of θ based on this sample and show that it indeed maximises the likelihood. [5]

SOLUTION. We have: [1-E]

$$L(\theta) = \prod_{i=1}^{n} f_X(x_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi b^2 \sigma^2}} e^{-\frac{(x_i - a\theta)^2}{2b^2\sigma_v^2}} = \frac{1}{(2\pi b^2\sigma^2)^{n/2}} e^{-\frac{\sum_{i=1}^{n}(x_i - a\theta)^2}{2b^2\sigma_v^2}}$$

Taking logarithms: [1-E]

$$\ell(\theta) = \log L(\theta) = -\frac{n}{2}\log(2\pi b^2\sigma^2) - \frac{1}{2b^2\sigma_v^2}\sum_{i=1}^{n}(x_i - a\theta)^2$$

Taking derivatives and setting to zero: [1-E]

$$\frac{\partial \ell(\theta)}{\partial \theta} = \frac{a}{b^2\sigma_v^2}\sum_{i=1}^{n}(x_i - a\theta) = 0$$

Rearranging and solving for θ: [1-E]

$$\widehat{\theta} = \frac{\frac{1}{n}\sum_{i=1}^{n} x_i}{a}$$

To show that this is indeed a maximum, we differentiate the derivative again:

$$\frac{\partial^2 \ell(\theta)}{\partial \theta^2} = \frac{-a^2 n}{b^2\sigma_v^2} < 0$$

Since the second derivative is always negative, every critical point must be a maximum (in fact, the function is concave). [1-E]

d) Is the maximum likelihood estimate computed in c) unbiased? Derive an expression for its variance. [7]

SOLUTION. The estimate is indeed unbiased. [1-E]

To verify this, note that: [1-E]

$$E(\widehat{\theta}) = \frac{\frac{1}{n}\sum_{i=1}^{n} E(X_i)}{a} = \frac{\frac{1}{n}\sum_{i=1}^{n} a\theta}{a} = \frac{\frac{1}{n} n a\theta}{a} = \theta$$

To compute the variance, we proceed as follows: [5-E]

$$\begin{aligned}
E\left(\widehat{\theta} - E(\widehat{\theta})\right)^2 &= E\left(\frac{1}{na}\sum_{i=1}^{n} X_i - \theta\right)^2 \\
&= E\left(\frac{1}{na}\sum_{i=1}^{n}(X_i - a\theta)\right)^2 \\
&\overset{(i)}{=} \frac{1}{n^2a^2}\sum_{i=1}^{n} E\left(X_i - a\theta\right)^2 \\
&\overset{(ii)}{=} \frac{1}{n^2a^2} n b^2\sigma_v^2 \\
&= \frac{b^2\sigma_v^2}{na^2}
\end{aligned}$$

where (i) follows from the fact that the X_i are i.i.d. with mean $a\theta$ and (ii) follows from the fact that X_i has variance $b^2\sigma_v^2$.